NYTU ANALYTICS

Architecture Document

VENDOR RISK ANALYSIS & PROCUREMENT INSIGHTS SOLUTION

SYSTEM ARCHITECTURE & TECHNICAL DESIGN OVERVIEW

Team:	|	**NYTU Analytics**
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Team Members:	|	**Natalia Flores** **Umaiza Farooque** **Yasmien Jemaledin** **Taylor Winston**

DOCUMENT OVERVIEW

This document describes the system architecture for the NYTU Analytics Vendor Risk Analysis solution. It outlines how raw procurement data flows through KNIME processing workflows into Tableau visualizations, delivering structured insights on vendor risk, spending concentration, and supplier diversity.

1. Solution Overview

This solution transforms raw procurement data into actionable insights related to vendor risk, spending concentration, and supplier diversity. The system integrates KNIME for data processing and Tableau for visualization, enabling structured analysis and informed procurement decision-making.

2. Architecture Flow

The diagram below illustrates the end-to-end data flow from raw input through processing to final business insights:

1. **Raw Procurement Data**
   2024–2025 CSV Files

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2. **KNIME Data Processing Workflow**
   Data cleaning, standardization & transformation

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3. **Vendor-Level Aggregation**
   GroupBy operations: total spend, transaction counts, avg unit price

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4. **Risk Classification Model**
   Rule-based classification: High / Medium / Low Risk

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5. **Processed Output Dataset**
   vendor_risk_output.csv + Pareto outputs

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6. **Tableau Dashboard Visualization**
   Interactive KPIs, charts, and SWAM analysis

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7 **Business Insights & Strategic Recommendations**
Actionable outputs for procurement decision-making

3. System Components

3.1 Data Sources

The input data consists of procurement datasets from 2024 and 2025. These datasets include vendor names, transaction amounts (Line Total), unit pricing, and purchasing activity across the organization.

▸ 3.2 Data Processing (KNIME)

- Combining multiple yearly datasets into a unified dataset
- Standardizing vendor names to eliminate duplicates and naming inconsistencies
- Cleaning and formatting numeric fields for accurate calculation
- Filtering and structuring the dataset for downstream consistency

This step ensures a reliable and consistent dataset for all downstream analysis and reporting.

▸ 3.3 Vendor-Level Aggregation

- Total vendor spend across the combined dataset
- Transaction counts per vendor
- Average unit price per vendor

After cleaning, the data is aggregated at the vendor level using GroupBy operations in KNIME. This aggregation enables comparison across vendors and forms the foundation for risk analysis.

▸ 3.4 Risk Classification Model

- High Risk — vendors with the highest total spend and greatest dependency impact
- Medium Risk — vendors with moderate spend and manageable exposure
- Low Risk — vendors with minimal spend and low operational dependency

A rule-based classification model is applied in KNIME to assign vendors into risk tiers. Risk is defined based on spending concentration and dependency; vendors with higher total spend are classified as higher risk due to their potential impact on operations if disrupted.

▸ 3.5 Data Outputs

- vendor_risk_output.csv — primary input for the Tableau dashboard
- Pareto-related outputs for cumulative spend analysis

The processed dataset is exported from KNIME and serves as the foundation for visualization and insight generation.

▸ 3.6 Data Visualization (Tableau)

- Total procurement spend (KPI summary)
- Top vendors by spend with associated risk classification
- Vendor spend concentration using Pareto analysis
- SWAM vs. Non-SWAM vendor spend comparison

Note: The SWAM analysis is based on the subset of data where SWAM classification was available.

4. Business Purpose

This architecture supports procurement decision-making by converting raw transaction data into structured, actionable insights. It enables organizations to:

- Identify high-risk vendors with significant spending concentration
- Understand and monitor spending patterns across the vendor base
- Reduce dependency risk through data-informed diversification
- Improve supplier diversity through SWAM vendor analysis

5. Scalability

Design Feature	Benefit
Reusable KNIME Workflow	Future procurement datasets can be processed without redesigning the pipeline
Automated Data Processing	Reduces manual effort and minimizes the risk of error when updating data
Tableau Live Connection	Dashboard updates automatically when new output data is refreshed

Modular Component Design	Individual components (cleaning, aggregation, classification) can be updated independently

6. Conclusion

This architecture provides a structured, repeatable, and scalable approach to vendor risk analysis. By integrating automated data processing in KNIME with interactive visualization in Tableau, the solution delivers both technical capability and measurable business value.

The modular design ensures that as procurement data grows and evolves, the system can adapt without requiring a full rebuild, enabling continuous, informed, and strategic procurement decision-making.

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